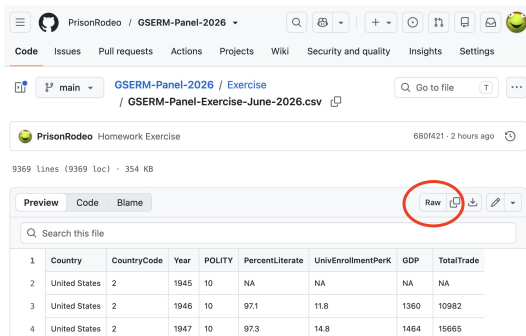


GSERM - St. Gallen 2025

Analyzing Panel Data

June 24, 2026

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/ GSERM-Panel-Exercise-June-2026.csv

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PrisonRodeo Homework Exercise 680f421 · 2 hours ago

9369 lines (9369 loc) · 354 KB

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	Country	CountryCode	Year	POLITY	PercentLiterate	UnivEnrollmentPerK	GDP	TotalTrade
1	United States	2	1945	10	NA	NA	NA	NA
2	United States	2	1946	10	97.1	11.8	1360	10982
3	United States	2	1947	10	97.3	14.8	1464	15665

Can also use (e.g.) `read.csv`:

```
> Data<-read.csv("https://raw.githubusercontent.com/PrisonRodeo/GSERM-Panel-2026/main/Exercise/GSERM-Panel-Exercise-June-2026.csv")
```

Correlated Random Effects (from yesterday...)

For the one-way unit effects model:

$$Y_{it} = \mathbf{X}_{it}\beta + \alpha_i + u_{it}$$

...standard “random effects” specifications require that:

$$\text{Cov}(\mathbf{X}_{it}, \alpha_i) = 0.$$

Recall that Mundlak (1978) proposed:

$$Y_{it} = \mathbf{X}_{it}\beta + \bar{\mathbf{X}}_i\gamma + u_{it}$$

which accounts for the fact that any correlation between (random) α_i and $\mathbf{X}$ occurs through the $\bar{\mathbf{X}}_i$ s...

Correlated Random Effects (continued)

We can think of this further by considering that the individual effect is (stochastically) related to the means of the $\mathbf{X}$ s:

$$\alpha_i = a + \bar{\mathbf{X}}_i \gamma + \varepsilon_i \quad (1)$$

Substituting that into our standard one-way model yields:

$$Y_{it} = \mathbf{X}_{it} \beta + \bar{\mathbf{X}}_i \gamma + \varepsilon_i + u_{it}$$

where ε_i is an (independent) unit-specific error component. This can be estimated by standard random effects methods, and Mundlak's specification test becomes a direct test for whether the *conventional* RE model is appropriate or not...

Mundlak's CREM

```
> MCREM<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents+PostColdWar+PGBetween+UPBetween+FRBetween+
  GDPBetween+NRRBetween+PCWBetween,data=smol,index=c("IS03","YearNumeric"),effect="individual",model="random",random.method = "walhus")

> summary(MCREM)
Oneway (individual) effect Random Effect Model
(Wallace-Hussain's transformation)

Call:
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
  FertilityRate + lnGDPPerCap + NaturalResourceRents + PostColdWar +
  PGBetween + UPBetween + FRBetween + GDPBetween + NRRBetween +
  PCWBetween, data = smol, effect = "individual", model = "random",
  random.method = "walhus", index = c("IS03", "YearNumeric"))

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Effects:
      var std.dev share
idiosyncratic 61.41    7.84 0.39
individual    95.61    9.78 0.61
theta:
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      0.727  0.877  0.888  0.881  0.888  0.888

Residuals:
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-32.98765 -5.32254 -0.52960 -0.00715  5.20319  53.81860

Coefficients:
              Estimate Std. Error z-value      Pr(>|z|)
(Intercept)    57.2306   12.3683    4.63 0.0000037064039 ***
PopGrowth       -0.1303    0.0806   -1.62      0.11
UrbanPopulation  0.3370    0.0183   18.40 < 2e-16 ***
FertilityRate   -1.5477    0.1549   -9.99 < 2e-16 ***
lnGDPPerCap     8.5685    0.2896   29.59 < 2e-16 ***
NaturalResourceRents 0.0898    0.0166    5.40 0.000000649079 ***
PostColdWar     6.7013    0.2862   23.42 < 2e-16 ***
PGBetween       -4.8688    1.0960   -4.44 0.000089009785 ***
UPBetween       -0.4037    0.0567   -7.12 0.0000000000011 ***
FRBetween        0.0207    1.1428    0.02    0.99
GDPBetween      -6.4288    1.1291   -5.69 0.0000000124273 ***
NRRBetween      -0.3921    0.0905   -4.33 0.0000148052468 ***
PCWBetween      -0.2002    5.3239   -0.04    0.97
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 1070000
Residual Sum of Squares: 506000
R-Squared: 0.527
Adj. R-Squared: 0.526
Chisq: 8978.36 on 12 DF, p-value: <2e-16
```

Mundlak's CREM: Testing

```
> linearHypothesis(MCREM,  
+ c("PGBetween = 0",  
+   "UPBetween = 0",  
+   "FRBetween = 0",  
+   "GDPBetween = 0",  
+   "NRRBetween = 0",  
+   "PCWBetween = 0"))
```

Linear hypothesis test:

```
PGBetween = 0  
UPBetween = 0  
FRBetween = 0  
GDPBetween = 0  
NRRBetween = 0  
PCWBetween = 0
```

Model 1: restricted model

Model 2: $\text{WomenBusLawIndex} \sim \text{PopGrowth} + \text{UrbanPopulation} + \text{FertilityRate} +$
 $\ln\text{GDPPerCap} + \text{NaturalResourceRents} + \text{PostColdWar} + \text{PGBetween} +$
 $\text{UPBetween} + \text{FRBetween} + \text{GDPBetween} + \text{NRRBetween} + \text{PCWBetween}$

```
      Res.Df Df Chisq Pr(>Chisq)  
1      8245  
2      8239  6   442    <2e-16 ***  
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

For the model of *WBLI*:

- The RE model's standard assumption that $\text{Cov}(\mathbf{X}_{it}, \alpha_i) = 0$ is unsupported by any test;
- Countries differ systematically in ways that are correlated with their long-run levels of $\mathbf{X}$, so
- The standard random effects model is inappropriate

BUT...

- The CREM (reported above) *is* appropriate, because the α_i s can be correlated with the $\mathbf{X}$ s.

FE and Mundlak Models

Table: Random Effects and Mundlak's CREM

	RE	MCREM
Population Growth	-0.190** (0.082)	-0.130 (0.081)
Urban Population	0.230*** (0.017)	0.340*** (0.018)
Fertility Rate	-1.900*** (0.150)	-1.500*** (0.150)
ln(GDP Per Capita)	7.000*** (0.270)	8.600*** (0.290)
Natural Resource Rents	0.058*** (0.017)	0.090*** (0.017)
Post-Cold War	7.900*** (0.280)	6.700*** (0.290)
Between-Country Population Growth		-4.900*** (1.100)
Between-Country Urban Population		-0.400*** (0.057)
Between-Country Fertility Rate		0.021 (1.100)
Between-Country ln(GDP Per Capita)		-6.400*** (1.100)
Between-Country Nat. Resource Rents		-0.390*** (0.091)
Between-Country Post-Cold War		-0.200 (5.300)
Constant	-8.700*** (2.500)	57.000*** (12.000)
Observations	8,252	8,252
R ²	0.500	0.530
Adjusted R ²	0.500	0.530
F Statistic	8,107.000***	8,978.000***

* p<0.1; ** p<0.05; *** p<0.01

Temporal Variation...

Generalized Least Squares Models

Start with a focus on residuals...

For:

$$Y_{it} = \mathbf{X}_{it}\beta + u_{it}$$

i.i.d. OLS u_{it} s require:

$$\mathbf{u}\mathbf{u}' \equiv \mathbf{\Omega} = \sigma^2\mathbf{I}$$

$$= \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix}$$

This means that within units:

- $\text{Var}(u_{it}) = \text{Var}(u_{is}) \forall t \neq s$ (temporal homoscedasticity)
- $\text{Cov}(u_{it}, u_{is}) = 0 \forall t \neq s$ (no within-unit autocorrelation)

and between units:

- $\text{Var}(u_{it}) = \text{Var}(u_{jt}) \forall i \neq j$ (cross-unit homoscedasticity)
- $\text{Cov}(u_{it}, u_{jt}) = 0 \forall i \neq j$ (no between-unit / spatial correlation)

Estimator:

$$\hat{\beta}_{GLS} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{Y}$$

with:

$$\widehat{V(\beta_{GLS})} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}$$

Two approaches:

- Use OLS $\hat{u}_{it}$ s to get $\hat{\Omega}$ (“feasible GLS” / “weighted least squares”)
- Use substantive knowledge about the data to structure Ω

The variance-covariance matrix is:

$$\begin{aligned}\text{Var}(\hat{\beta}_{WLS}) &= \sigma^2(\mathbf{X}'\mathbf{W}^{-1}\mathbf{X})^{-1} \\ &\equiv (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\end{aligned}$$

A common case is:

$$\text{Var}(u_i) = \sigma^2 \frac{1}{N_i}$$

where N_i is the number of observations upon which (aggregate) observation i is based.

“Robust” Variance Estimators

Recall that, if $\sigma_i^2 \neq \sigma_j^2 \forall i \neq j$,

$$\begin{aligned}\text{Var}(\beta_{\text{Het.}}) &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \mathbf{Q} (\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

where $\mathbf{Q} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})$ and $\mathbf{\Omega} = \sigma^2\mathbf{W}$.

We can rewrite $\mathbf{Q}$ as:

$$\begin{aligned}\mathbf{Q} &= \sigma^2(\mathbf{X}'\mathbf{W}^{-1}\mathbf{X}) \\ &= \sum_{i=1}^N \sigma_i^2 \mathbf{x}_i \mathbf{x}_i'\end{aligned}$$

Estimate $\hat{\mathbf{Q}}$ as:

$$\hat{\mathbf{Q}} = \sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i'$$

Yields:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta})_{\text{Robust}} &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\hat{\mathbf{Q}}^{-1}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1} \left[\mathbf{X}' \left(\sum_{i=1}^N \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i' \right)^{-1} \mathbf{X} \right] (\mathbf{X}'\mathbf{X})^{-1} \end{aligned}$$

“Robust” VCV estimates:

- are heteroscedasticity-consistent, but
- are biased in small samples, and
- are less efficient than “naive” estimates when $\text{Var}(u) = \sigma^2 \mathbf{I}$.

In practice, they’ve become the “default;” but it’s important to understand their limitations

“Clustering”

Huber / White

?????????

WLS / GLS

I know very little
about my error
variances...

I know a great
deal about my
error variances...

A common case:

$$Y_{ij} = \mathbf{X}_{ij}\beta + u_{ij}$$

with

$$\sigma_{ij}^2 = \sigma_{ik}^2.$$

“Robust, clustered” estimator:

$$\widehat{\text{Var}}(\beta)_{\text{Clustered}} = (\mathbf{X}'\mathbf{X})^{-1} \left\{ \mathbf{X}' \left[\sum_{i=1}^N \left(\sum_{j=1}^{n_j} \hat{u}_{ij}^2 \mathbf{X}_{ij} \mathbf{X}_{ij}' \right) \right]^{-1} \mathbf{X} \right\} (\mathbf{X}'\mathbf{X})^{-1}$$

Robust / Clustered SEs: A Simulation

```
> set.seed(3844469)
> X <- rnorm(10)
> Y <- 1 + X + rnorm(10)
> df10 <- data.frame(ID=seq(1:10),X=X,Y=Y)

> fit10 <- feols(Y~X,data=df10)
> summary(fit10)

OLS estimation, Dep. Var.: Y
Observations: 10
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9536	0.3114	3.062	0.015534 *
X	0.5893	0.2907	2.027	0.077193 .

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.880655  Adj. R2: 0.256783

> fit10robust <- feols(Y~X,data=df10,vcov="hetero")
> summary(fit10robust)

OLS estimation, Dep. Var.: Y
Observations: 10
Standard-errors: Heteroskedasticity-robust

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9536	0.3147	3.030	0.016298 *
X	0.5893	0.2850	2.067	0.072517 .

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.880655  Adj. R2: 0.256783
```

Robust / Clustered SEs: A Simulation (continued)

```
> # "Clone" each observation 100 times:
>
> df1K <- df10[rep(seq_len(nrow(df10)),each=100),]
> df1K <- pdata.frame(df1K, index="ID")

> fit1K <- feols(Y~X,data=df1K)
> summary(fit1K)

OLS estimation, Dep. Var.: Y
Observations: 1,000
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9536	0.02788	34.20	< 2.2e-16 ***
X	0.5893	0.02603	22.64	< 2.2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.880655  Adj. R2: 0.3387

> # Robust, clustered SEs:
>
> fit1Krobust <- feols(Y~X,data=df1K,cluster="ID")
> summary(fit1Krobust)

OLS estimation, Dep. Var.: Y
Observations: 1,000
Standard-errors: Clustered (ID)

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9536	0.2968	3.213	0.010613 *
X	0.5893	0.2689	2.192	0.056090 .

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.880655  Adj. R2: 0.3387
```

Example:

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

$$u_t = \rho u_{t-1} + e_t$$

with $e_t \sim i.i.d. N(0, \sigma_u^2)$ and $\rho \in [-1, 1]$ (typically).

→ “First-order autoregressive” [“AR(1)”] errors.

Serially Correlated Errors and OLS

Detection

- *Plot* of residuals vs. lagged residuals
- *Runs* test (Geary test)
- Durbin-Watson d
 - Calculated as:

$$d = \frac{\sum_{t=2}^N (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^N \hat{u}_t^2}$$

- Non-standard distribution ($d \in [0, 4]$)
- Null: No autocorrelation
- Only detects first-order autocorrelation

Serially Correlated Errors and OLS

What to do about it?

- GLS, incorporating ρ / $\hat{\rho}$ into the equation
- *First-difference* models (regressing changes of Y on changes of $\mathbf{X}$)
- Cochrane-Orcutt / Prais-Winsten:
 1. Estimate the basic equation via OLS, and obtain residuals
 2. Use the residuals to consistently estimate $\hat{\rho}$ (i.e. the empirical correlation between u_t and u_{t-1})
 3. Use this estimate of $\hat{\rho}$ to estimate the *difference equation*:

$$(Y_t - \rho Y_{t-1}) = \beta_0(1 - \rho) + \beta_1(X_t - \rho X_{t-1}) + (u_t - \rho u_{t-1})$$

4. Save the residuals, and use them to estimate $\hat{\rho}$ again
5. Repeat this process until successive estimates of $\hat{\rho}$ differ by a very small amount

Running Example Redux

The World Development Indicators:

- Cross-national country-level time series data
- $N = 215$ countries, $T = 65$ years (1960-2024) + missingness
- Full descriptions are listed in the Github repo [here](#)

Regression model:

$$\text{WBLI}_{it} = \beta_0 + \beta_1 \text{Population Growth}_{it} + \beta_2 \text{Urban Population}_{it} + \beta_3 \text{Fertility Rate}_{it} + \beta_4 \ln(\text{GDP Per Capita})_{it} + \beta_5 \text{Natural Resource Rents}_{it} + \beta_6 \text{Post-Cold War}_t + u_{it}$$

Descriptive Statistics:

	vars	n	mean	sd	median	min	max	range	skew	kurtosis	se
WomenBusLawIndex	1	8252	60.03	18.85	60.60	17.50	100.00	82.50	-0.02	-0.66	0.21
PopGrowth	2	8252	1.67	1.67	1.64	-27.47	21.70	49.17	-0.09	30.12	0.02
UrbanPopulation	3	8252	52.48	23.88	53.04	3.01	100.00	96.99	0.03	-1.05	0.26
FertilityRate	4	8252	3.59	1.89	3.06	0.77	8.61	7.83	0.53	-1.01	0.02
NaturalResourceRents	5	8252	7.28	11.18	2.47	0.00	88.59	88.59	2.49	7.40	0.12
PostColdWar	6	8252	0.70	0.46	1.00	0.00	1.00	1.00	-0.87	-1.24	0.01
lnGDPPerCap	7	8252	8.34	1.46	8.25	4.93	11.67	6.74	0.11	-0.89	0.02

A Digression: Rescaling Covariates

A la [Gelman \(2008\)](#) (and an updated blog post [here](#)):

- Continuous = divide by one standard deviation
- Binary = recode to $\{-1, 1\}$

Doing this yields:

	vars	n	mean	sd	median	min	max	range	skew	kurtosis	se
WomenBusLawIndex	1	8252	60.03	18.85	60.60	17.50	100.00	82.50	-0.02	-0.66	0.21
PopGrowth	2	8252	1.00	1.00	0.98	-16.42	12.97	29.39	-0.09	30.12	0.01
UrbanPopulation	3	8252	2.20	1.00	2.22	0.13	4.19	4.06	0.03	-1.05	0.01
FertilityRate	4	8252	1.90	1.00	1.62	0.41	4.56	4.15	0.53	-1.01	0.01
NaturalResourceRents	5	8252	0.65	1.00	0.22	0.00	7.92	7.92	2.49	7.40	0.01
PostColdWar	6	8252	0.40	0.92	1.00	-1.00	1.00	2.00	-0.87	-1.24	0.01
lnGDPPerCap	7	8252	5.72	1.00	5.66	3.38	8.01	4.63	0.11	-0.89	0.01

How Much Autocorrelation in Y ?

Note that:

$$d = 2(1 - \rho)$$

which means that we can calculate:

$$\rho = 1 - \frac{d}{2}.$$

So:

```
> WI<-pdwtest(WomenBusLawIndex~1,data=smol)
> WI
```

Durbin-Watson test for serial correlation in panel models

```
data:  WomenBusLawIndex ~ 1
DW = 0.097, p-value <2e-16
alternative hypothesis: serial correlation in idiosyncratic errors
```

```
> print(paste("Rho =",round(1 - (WI$statistic/2),3)))
[1] "Rho = 0.951"
```

How Much Autocorrelation in **X**?

Table: WDI Data - Autocorrelation in the Predictors

Variable	Rho
Population Growth	0.779
Urban Population	0.975
Fertility Rate	0.965
GDP Per Capita	0.977
Natural Resource Rents	0.914
Post Cold War	0.913

Baseline Model: OLS (+ D-W Test)

```
> OLS<-plm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPERCap+NaturalResourceRents+
+          PostColdWar,data=smol,model="pooling")
> summary(OLS)
```

Call:

```
plm(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
     FertilityRate + lnGDPPERCap + NaturalResourceRents + PostColdWar,
     data = smol, model = "pooling")
```

Unbalanced Panel: n = 188, T = 8-51, N = 8252

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-78.29	-8.62	1.01	9.27	45.19

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	56.904	1.707	33.34	< 2e-16 ***
PopGrowth	-2.985	0.195	-15.31	< 2e-16 ***
UrbanPopulation	-1.787	0.248	-7.22	5.7e-13 ***
FertilityRate	-5.641	0.292	-19.31	< 2e-16 ***
lnGDPPERCap	3.729	0.274	13.62	< 2e-16 ***
NaturalResourceRents	-3.921	0.168	-23.32	< 2e-16 ***
PostColdWar	4.941	0.183	27.01	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 2930000

Residual Sum of Squares: 1490000

R-Squared: 0.493

Adj. R-Squared: 0.493

F-statistic: 1337.18 on 6 and 8245 DF, p-value: <2e-16

```
> pdwtest(OLS)
```

Durbin-Watson test for serial correlation in panel models

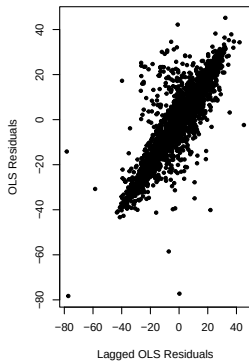
data: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + ...

DW = 0.14, p-value <2e-16

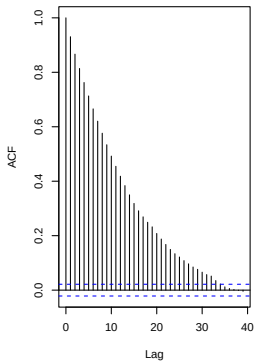
alternative hypothesis: serial correlation in idiosyncratic errors

Residual Autocorrelation?

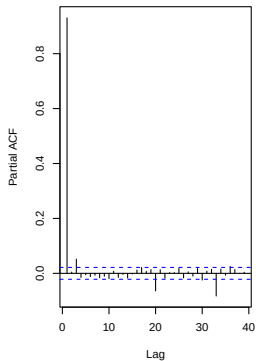
Residuals vs. Lagged Residuals



ACF of OLS Residuals

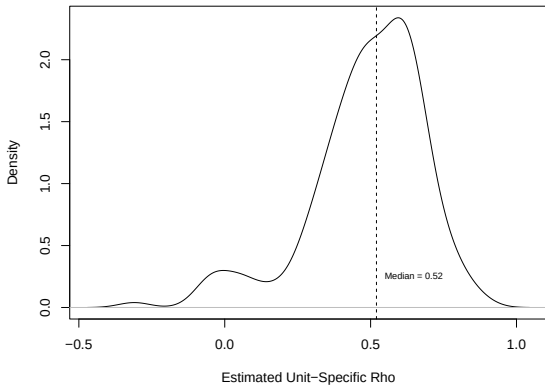


PACF of OLS Residuals



Unit-Specific Autocorrelation...

Fit $N = 188$ country-specific regressions, and examine the $\hat{\rho}_i$ s...



Another Model: Prais-Winsten

```
> PraisWinsten<-panelAR(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents+
+       PostColdWar, data=smol,panelVar="IS03",timeVar="Year",autoCorr="ar1",panelCorrMethod="none",rho.na.rm=TRUE)
```

```
> summary(PraisWinsten)
```

Panel Regression with AR(1) Prais-Winsten correction and homoskedastic variance

Unbalanced Panel Design:

```
Total obs.:      8252 Avg obs. per panel 43.894
Number of panels: 188 Max obs. per panel 51
Number of times:  51  Min obs. per panel  8
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	69.1076	2.9188	23.68	<2e-16 ***
PopGrowth	0.0330	0.0463	0.71	0.4765
UrbanPopulation	0.5926	0.5660	1.05	0.2951
FertilityRate	-9.4295	0.4262	-22.12	<2e-16 ***
lnGDPPerCap	1.4482	0.4903	2.95	0.0031 **
NaturalResourceRents	-0.1330	0.0842	-1.58	0.1142
PostColdWar	0.3071	0.1042	2.95	0.0032 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-squared: 0.2889

Wald statistic: 972.7296, Pr(>Chisq(6)): 0

```
> PraisWinsten$panelStructure$rho
[1] 0.9559
```

WBLI Regressions		
	OLS	Prais-Winsten
Intercept	56.90*	69.10*
	(1.71)	(2.92)
Population Growth	-2.98*	0.033
	(0.19)	(0.046)
Urban Population	-1.79*	0.593
	(0.25)	(0.566)
Fertility Rate	-5.64*	-9.43*
	(0.29)	(0.43)
ln(GDP Per Capita)	3.73*	1.44*
	(0.27)	(0.49)
Natural Resource Rents	-3.92*	-0.13
	(0.17)	(0.08)
Post Cold War	4.94*	0.31*
	(0.18)	(0.10)
R ²	0.49	0.29
Adj. R ²	0.49	0.29
Num. obs.	8252	8252

Variables are standardized a la Gelman (2009). * $p < 0.05$

Some Panel Data Challenges

Consider the error terms in the model:

$$Y_{it} = \mathbf{X}_{it}\beta + u_{it}$$

Issues:

<u>In Words:</u>	<u>In a Formula:</u>
<u>Variances:</u>	
Unit-Wise Heteroscedasticity	$\text{Var}(u_{it}) \neq \text{Var}(u_{jt})$
Temporal Heteroscedasticity	$\text{Var}(u_{it}) \neq \text{Var}(u_{is})$
<u>Covariances:</u>	
Contemporary Cross-Unit Correlation	$\text{Cov}(u_{it}, u_{jt}) \neq 0$
Within-Unit Serial Correlation	$\text{Cov}(u_{it}, u_{is}) \neq 0$
Non-Contemporaneous Cross-Unit Correlation	$\text{Cov}(u_{it}, u_{js}) \neq 0$

Parks' (1967) Approach

Assume:

- $\text{Var}(u_{it}, u_{jt}) = \sigma^2$ or σ_i^2 (Common or unit-specific error variances)
- $\text{Var}(u_{it}) = \text{Var}(u_{is}) \forall t \neq s$ (Temporal homoscedasticity)
- $\text{Cov}(u_{it}, u_{jt}) = \sigma_{ij} \forall i \neq j$ (Pairwise contemporaneous cross-unit correlation)
- $\text{Cov}(u_{it}, u_{is}) = \rho$ or ρ_i (Common or unit-specific temporal correlation)
- $\text{Cov}(u_{it}, u_{js}) = 0 \forall i \neq j, t \neq s$ (No non-contemporaneous cross-unit correlation)

(Beck & Katz (1995): “panel error assumptions”).

Then:

1. Use OLS to generate $\hat{u}s \rightarrow \hat{\rho} (\rightarrow \hat{\Omega})$,
2. Use $\hat{\rho}$ for Prais-Winsten.

This method was widely used prior to Beck & Katz (1995)...

The variance-covariance matrix is:

$$\mathbf{\Omega} = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & \Sigma & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Sigma \end{pmatrix} = \Sigma \otimes \mathbf{I}_T$$

where

$$\Sigma_{N \times N} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1N} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1N} & \sigma_{2N} & \cdots & \sigma_N^2 \end{pmatrix}$$

Means:

- $\frac{N(N-1)}{2}$ distinct contemporaneous covariances σ_{ij} ,
- NT observations,
- $\rightarrow 2T/(N+1)$ observations per $\hat{\sigma}$

From PROC PANEL in SAS:

Standard Corrections

For the PARKS option, the first-order autocorrelation coefficient must be estimated for each cross section. Let ρ be the $N \times 1$ vector of true parameters and $R = (r_1, \dots, r_N)'$ be the corresponding vector of estimates. Then, to ensure that only range-preserving estimates are used in PROC PANEL, the following modification for R is made:

$$r_i = \begin{cases} r_i & \text{if } |r_i| < 1 \\ \max(.95, r_{\max}) & \text{if } r_i \geq 1 \\ \min(-.95, r_{\min}) & \text{if } r_i \leq -1 \end{cases}$$

where

$$r_{\max} = \begin{cases} 0 & \text{if } r_i < 0 \text{ or } r_i \geq 1 \quad \forall i \\ \max_j [r_j : 0 \leq r_j < 1] & \text{otherwise} \end{cases}$$

and

$$r_{\min} = \begin{cases} 0 & \text{if } r_i > 0 \text{ or } r_i \leq -1 \quad \forall i \\ \max_j [r_j : -1 < r_j \leq 0] & \text{otherwise} \end{cases}$$

Whenever this correction is made, a warning message is printed.

Panel-Corrected Standard Errors

Key to PCSEs:

$$\hat{\sigma}_{ij} = \frac{\sum_{t=1}^T \hat{u}_{it} \hat{u}_{jt}}{T}$$

Define:

$$\mathbf{U}_{T \times N} = \begin{pmatrix} \hat{u}_{11} & \hat{u}_{21} & \cdots & \hat{u}_{N1} \\ \hat{u}_{12} & \hat{u}_{22} & \cdots & \hat{u}_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{u}_{1T} & \hat{u}_{2T} & \cdots & \hat{u}_{NT} \end{pmatrix}$$

$$\hat{\Sigma} = \frac{(\mathbf{U}'\mathbf{U})}{T}$$

$$\hat{\Omega}_{PCSE} = \frac{(\mathbf{U}'\mathbf{U})}{T} \otimes \mathbf{I}_T$$

Panel-Corrected Standard Errors

Correct formula:

$$\text{Cov}(\hat{\beta}_{PCSE}) = (\mathbf{X}'\mathbf{X})^{-1}[\mathbf{X}'\hat{\Omega}_{PCSE}\mathbf{X}](\mathbf{X}'\mathbf{X})^{-1}$$

General Issues:

- PCSEs do not fix unit-level heterogeneity (a la “fixed” / “random” effects)
- They also do not deal with dynamics
- They depend critically on the “panel data assumptions” of Park / Beck & Katz

Panel Assumptions and Numbers of Parameters

Panel Assumptions	No AR(1)	Common $\hat{\rho}$	Separate $\hat{\rho}_i$ s
$\sigma_i^2 = \sigma^2, \text{Cov}(\sigma_{it}, \sigma_{jt}) = 0$	$k + 1$	$k + 2$	$k + N + 1$
$\sigma_i^2 \neq \sigma^2, \text{Cov}(\sigma_{it}, \sigma_{jt}) = 0$	$k + N$	$k + N + 1$	$k + 2N$
$\sigma_i^2 \neq \sigma^2, \text{Cov}(\sigma_{it}, \sigma_{jt}) \neq 0$	$\frac{N(N-1)}{2} + k + N$	$\frac{N(N-1)}{2} + k + N + 1$	$\frac{N(N-1)}{2} + k + 2N$

Example: GLS with Homoscedastic AR(1) Errors

```
> GLS<-gas(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents+
+ PostColdWar,data=smol,correlation=corAR1(form=~1|ISO3),na.action="na.omit")
>
> summary(GLS)
Generalized least squares fit by REML
Model: WomenBusLawIndex ~ PopGrowth + UrbanPopulation + FertilityRate + lnGDPPerCap + NaturalResourceRents + PostColdWar
Data: smol
    AIC      BIC logLik
38705 38768 -19344

Correlation Structure: AR(1)
Formula: ~1 | ISO3
Parameter estimate(s):
    Phi
0.9896

Coefficients:
              Value Std.Error t-value p-value
(Intercept)   50.45    3.879   13.008  0.0000
PopGrowth      0.04    0.044    0.924  0.3554
UrbanPopulation 6.06    0.870    6.965  0.0000
FertilityRate -6.32    0.522  -12.104  0.0000
lnGDPPerCap    1.70    0.585    2.901  0.0037
NaturalResourceRents 0.14  0.077    1.885  0.0595
PostColdWar    0.09    0.099    0.866  0.3866

Correlation:
              (Intr) PpGrwt UrbnPp FrtltR lGDPPC NtrlRR
PopGrowth      0.033
UrbanPopulation -0.326  0.006
FertilityRate  -0.520 -0.050  0.408
lnGDPPerCap    -0.725 -0.038 -0.312  0.068
NaturalResourceRents 0.010 -0.008  0.024  0.032 -0.049
PostColdWar    -0.003  0.028 -0.041  0.063  0.001 -0.020

Standardized residuals:
      Min      Q1      Med      Q3      Max
-3.391517 -0.568108 -0.001919  0.527836  2.089031

Residual standard error: 16.76
Degrees of freedom: 8252 total; 8245 residual
```


Example: PCSEs

```
> PCSE<-panelAR(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents+
+               PostColdWar,data=smol,panelVar="ISO3",timeVar="Year",autoCorr="ar1",
+               panelCorrMethod="pcse",rho.na.rm=TRUE)
```

```
> summary(PCSE)
```

Panel Regression with AR(1) Prais-Winsten correction and panel-corrected standard errors

Unbalanced Panel Design:

```
Total obs.:      8252 Avg obs. per panel 43.894
Number of panels: 188 Max obs. per panel 51
Number of times:  51  Min obs. per panel  8
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	69.1076	4.3401	15.92	<2e-16 ***
PopGrowth	0.0330	0.0516	0.64	0.523
UrbanPopulation	0.5926	0.6587	0.90	0.368
FertilityRate	-9.4295	0.6868	-13.73	<2e-16 ***
lnGDPPerCap	1.4482	0.6714	2.16	0.031 *
NaturalResourceRents	-0.1330	0.1288	-1.03	0.302
PostColdWar	0.3071	0.2805	1.09	0.274

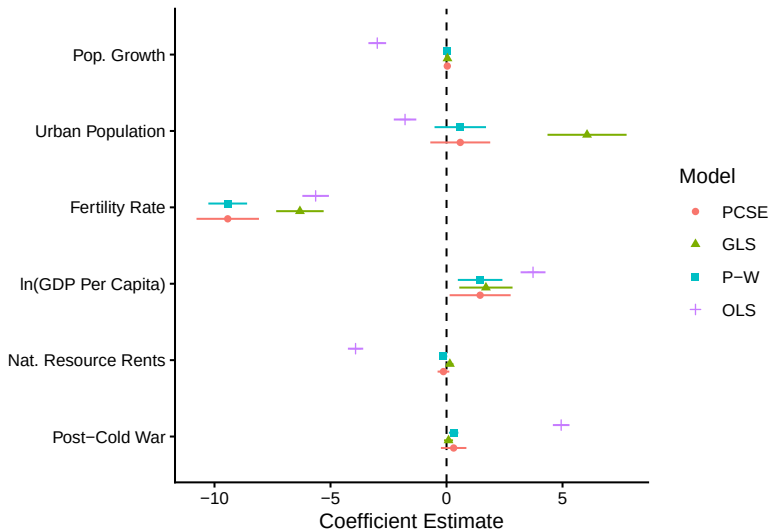
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-squared: 0.2889

Wald statistic: 318.036, Pr(>Chisq(6)): 0

```
> PCSE$panelStructure$rho
[1] 0.9559
```

Estimate Comparisons



Dynamics!

Time Series: Stationarity

Stationarity: A constant d.g.p. over time.¹

Mean stationarity:

$$E(Y_t) = \mu \quad \forall t$$

Variance stationarity:

$$\text{Var}(Y_t) = E[(Y_t - \mu)^2] \equiv \sigma_Y^2 \quad \forall t$$

Covariance stationarity:

$$\text{Cov}(Y_t, Y_{t-s}) = E[(Y_t - \mu)(Y_{t-s} - \mu)] = \gamma_s \quad \forall s$$

¹A stricter form of stationarity requires that the joint probability distribution (in other words, *all* the moments) of series of observations $\{Y_1, Y_2, \dots, Y_t\}$ is the same as that for $\{Y_{1+s}, Y_{2+s}, \dots, Y_{t+s}\}$ for all t and s .

The “ARIMA” Approach

“ARIMA” = *Autoregressive Integrated Moving Average*...

A (first-order) integrated series (“random walk”) is:

$$Y_t = Y_{t-1} + u_t, \quad u_t \sim i.i.d.(0, \sigma_u^2)$$

...a/k/a a “random walk”:

$$\begin{aligned} Y_t &= Y_{t-2} + u_{t-1} + u_t \\ &= Y_{t-3} + u_{t-2} + u_{t-1} + u_t \\ &= \sum_{t=0}^T u_t \end{aligned}$$

I(1) Series Properties

I(1) series are not stationary.

Variance:

$$\text{Var}(Y_t) \equiv E(Y_t)^2 = t\sigma^2$$

Autocovariance:

$$\text{Cov}(Y_t, Y_{t-s}) = |t - s|\sigma^2.$$

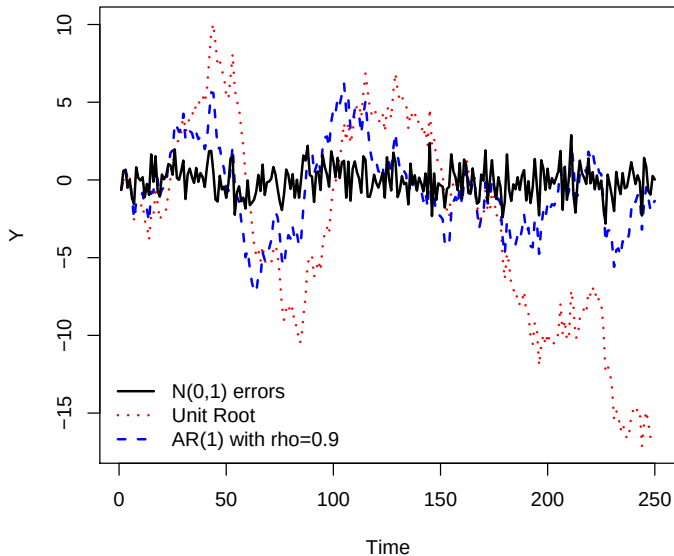
Both depend on t ...

I(1) series (continued)

More generally:

- $|\rho| > 1$
 - Series is nonstationary / *explosive*
 - Past shocks have a greater impact than current ones
 - Uncommon
- $|\rho| < 1$
 - *Stationary* series
 - Effects of shocks die out exponentially according to ρ
 - Is mean-reverting
- $|\rho| = 1$
 - Nonstationary series
 - Shocks persist at full force
 - Not mean-reverting; variance increases with t

Time Series Types, Illustrated



I(1) Series: Differencing

For an I(1) series:

$$Y_t - Y_{t-1} = u_t$$

which we often write in terms of the difference operator Δ (or sometimes ∇):

$$\Delta Y_t = u_t$$

The differenced series is just the (stationary, ergodic) white-noise process u_t .

Unit Root Tests Review: Dickey-Fuller

Two steps:

- Estimate $Y_t = \rho Y_{t-1} + u_t$,
- test the hypothesis that $\hat{\rho} = 1$, *but*
- this requires that the u s are uncorrelated.

But suppose:

$$\Delta Y_t = \sum_{i=1}^p d_i \Delta Y_{t-i} + u_t$$

which yields

$$Y_t = Y_{t-1} + \sum_{i=1}^p d_i \Delta Y_{t-i} + u_t.$$

D.F. tests will be incorrect.

Augmented Dickey-Fuller Tests:

- Estimate

$$\Delta Y_t = \rho Y_{t-1} + \sum_{i=1}^p d_i \Delta Y_{t-i} + u_t$$

- Test $\hat{\rho} = 0$ (vs. alternative that $\hat{\rho} < 0$)

Phillips-Perron Tests:

- Estimate:

$$\Delta Y_t = \alpha + \rho Y_{t-1} + u_t$$

- Calculate modified test statistics (Z_ρ and Z_t)
- Test $\hat{\rho} = 0$

Issues with Unit Roots in Panel Data

- Short series + Asymptotic tests $\rightarrow$ “borrow strength”
- Requires uniform unit roots across cross-sectional units
- Many tests require balanced panels...
- Various alternatives:
 - Maddala and Wu (1999)
 - Hadri (2000)
 - Levin, Lin and Chu (2002)
 - Im, Pesaran, and Shin (2003)
- What to do?
 - Difference the data...
 - Error-correction models

Panel Unit Root Tests: R

```
[data wrangling...]
```

```
> purtest(WBLI.W,exo="trend",test="levinlin",pmax=2)
```

```
Levin-Lin-Chu Unit-Root Test (ex. var.: Individual Intercepts and Trend)
```

```
data: WBLI.W
```

```
z = -1.7e+16, p-value <2e-16
```

```
alternative hypothesis: stationarity
```

```
> purtest(WBLI.W,exo="trend",test="hadri",pmax=2)
```

```
Hadri Test (ex. var.: Individual Intercepts and Trend) (Heterosked. Consistent)
```

```
data: WBLI.W
```

```
z = 201, p-value <2e-16
```

```
alternative hypothesis: at least one series has a unit root
```

```
> purtest(WBLI.W,exo="trend",test="madwu",pmax=2)
```

```
Maddala-Wu Unit-Root Test (ex. var.: Individual Intercepts and Trend)
```

```
data: WBLI.W
```

```
chisq = 296, df = 380, p-value = 1
```

```
alternative hypothesis: stationarity
```

```
> purtest(WBLI.W,exo="trend",test="ips",pmax=2)
```

```
Im-Pesaran-Shin Unit-Root Test (ex. var.: Individual Intercepts and Trend)
```

```
data: WBLI.W
```

```
Wtbar = -2.7e+15, p-value <2e-16
```

```
alternative hypothesis: stationarity
```

Table: Panel Unit Root Tests: WBRI

	Test	Alternative	Statistic	Estimate	P-Value
1	Levin-Lin-Chu	stationarity	z	$<-16527b$	0
2	Hadri	at least one series has a unit root	z	201.362	0
3	Maddala-Wu	stationarity	χ^2	296.481	0.9994
4	Im-Pesaran-Shin	stationarity	$\bar{W}_t$	$<-2670b$	0

Note: All assume individual intercepts and trends.

“Lagged dependent variable”:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\beta_{LDV} + \epsilon_{it}$$

If ϵ_{it} is perfect, then:

- $\hat{\beta}_{LDV}$ is biased (but consistent),
- $O(\text{bias}) = \frac{-1+3\beta_{LDV}}{T}$

If ϵ_{it} is autocorrelated...

- $\hat{\beta}_{LDV}$ is biased and inconsistent
- IV is one (bad) option...

Lagged Y s and GLS-ARMA

Can rewrite:

$$\begin{aligned}Y_{it} &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + u_{it} \\ u_{it} &= \phi u_{it-1} + \eta_{it}\end{aligned}$$

as

$$\begin{aligned}Y_{it} &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \phi u_{it-1} + \eta_{it} \\ &= \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \phi(Y_{it-1} - \mathbf{X}_{it-1}\boldsymbol{\beta}_{AR}) + \eta_{it} \\ &= \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta}_{AR} + \mathbf{X}_{it-1}\psi + \eta_{it}\end{aligned}$$

where $\psi = \phi\boldsymbol{\beta}_{AR}$ and $\psi = 0$ (by assumption).

Lagged Y s and World Domination

In:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\beta_{LDV} + \epsilon_{it}$$

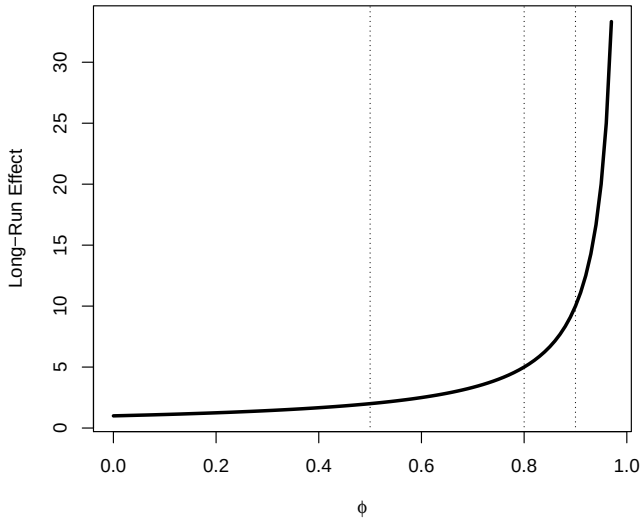
Achen: Bias “deflates” $\hat{\beta}_{LDV}$ relative to $\hat{\phi}$, “suppress” the effects of $\mathbf{X}$...

Keele & Kelly (2006):

- Contingent on ϵ s having autocorrelation
- Key: In LDV, *long-run impact of a unit change in X is:*

$$\hat{\beta}_{LR} = \frac{\hat{\beta}_{LDV}}{1 - \hat{\phi}}$$

Long-Run Impact for $\hat{\beta} = 1$



Lagged Y s and Unit Effects

Consider:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta} + \alpha_i + u_{it}.$$

If we omit the unit effects, we have:

$$Y_{it} = \phi Y_{it-1} + \mathbf{X}_{it}\boldsymbol{\beta} + u_{it}^*$$

with

$$u_{it}^* = \alpha_i + u_{it}$$

Lagging yields:

$$Y_{it-1} = \phi Y_{it-2} + \mathbf{X}_{it-1}\boldsymbol{\beta} + \alpha_i + u_{it-1}$$

which means

$$\text{Cov}(Y_{it-1}, u_{it}^*) \neq 0. \rightarrow \text{bias in } \hat{\phi}, \hat{\boldsymbol{\beta}}$$

Bias in $\hat{\phi}$ is

- toward zero when $\phi > 0$,
- increasing in ϕ .

Including unit effects still yields bias in $\hat{\phi}$ of $O(\frac{1}{T})$, and bias in $\hat{\beta}$.

Solutions:

- Difference/GMM estimation
- Bias correction approaches

First Difference Estimation

$$\begin{aligned}Y_{it} - Y_{it-1} &= \phi(Y_{it-1} - Y_{it-2}) + (\mathbf{X}_{it} - \mathbf{X}_{it-1})\beta + (\alpha_i - \alpha_i) + (u_{it} - u_{it-1}) \\ \Delta Y_{it} &= \phi \Delta Y_{it-1} + \Delta \mathbf{X}_{it}\beta + \Delta u_{it}\end{aligned}$$

Anderson/Hsiao: If $\nexists$ autocorrelation, then use ΔY_{it-2} or Y_{it-2} as instruments for ΔY_{it-1} ...

- Consistent in theory,
- in practice, the former is preferred, and both have issues if ϕ is high;
- both are inefficient.

Arellano & Bond: Use *all* lags of Y_{it} and $\mathbf{X}_{it}$ from $t - 2$ and before.

- “Good” estimates, better as $T \rightarrow \infty$,
- Easy to handle higher-order lags of Y ,
- Easy software (plm in R , xtabond in Stata).
- Model *is* fixed effects...
- $\mathbf{Z}_i$ has $T - p - 1$ rows, $\sum_{i=p}^{T-2} i$ columns $\rightarrow$ difficulty of estimation declines in p , grows in T .

Kiviet (1995, 1999; Bun and Kiviet 2003; Bruno 2005a,b): Derive the bias in $\hat{\phi}$ and $\hat{\beta}$, then correct it...

- More accurate than the instrumental-variables/GMM estimators of A&H/A&B...
- ...especially when T is small; but not as T gets reasonably large ($T \approx 20$)

Some Dynamic Models

	OLS	Lagged Y	First Difference	Fixed Effects	FE + Lagged Y	Anderson-Hsaio
(Intercept)	56.904 (1.707)	2.115 (0.320)	0.657 (0.037)			-0.158 (0.352)
Population Growth	-2.985 (0.195)	-0.028 (0.035)	0.043 (0.043)	-0.218 (0.135)	-0.041 (0.041)	-0.014 (0.191)
Urban Population	-1.787 (0.248)	0.038 (0.044)	0.014 (1.218)	8.047 (0.437)	0.339 (0.137)	-0.740 (36.757)
Fertility Rate	-5.641 (0.292)	-0.206 (0.053)	-0.571 (0.602)	-2.920 (0.292)	-0.508 (0.090)	-1.955 (14.391)
ln(GDP Per Capita)	3.729 (0.274)	-0.067 (0.049)	0.066 (0.627)	12.486 (0.422)	0.227 (0.136)	-2.945 (8.568)
Natural Resource Rents	-3.921 (0.168)	-0.102 (0.031)	0.213 (0.075)	1.005 (0.186)	0.013 (0.057)	-0.007 (0.752)
Post-Cold War	4.941 (0.183)	0.111 (0.034)	-0.143 (0.098)	3.351 (0.143)	0.157 (0.044)	0.508 (1.283)
Lagged WBLI		0.988 (0.002)			0.954 (0.003)	1.206 (0.324)
Num.Obs.	8252	8053	8064	8252	8053	7855
R2	0.493	0.985	0.002	0.520	0.957	0.003
R2 Adj.	0.493	0.985	0.001	0.508	0.956	0.003
Log.Lik.		-18 264.656				
RMSE	13.42	2.34	2.37	7.74	2.29	3.58

Anderson-Hsiao, Arellano-Bond, etc.

In R:

- Anderson-Hsiao can be fit using `lm` or (more easily) `plm` in the `plm` package
- Arellano-Bond is most easily fit using `pgmm` (“panel gmm”) in the `plm` package
- See Cribari and Millo (2018, Chapter 7) for statistics + code details
- [This post](#) by Guillermo Corredor is also useful...

Stata:

- `xtabond` / `xtdpdsys` / `xtdpd` fit both A-H and A-B / Blundell-Bond models (among others)
- [This](#) is also a good (slightly dated) reference

Another Approach: Orthogonalization

Note: We're rarely substantively interested in the fixed effects $\hat{\alpha}$...

- $\rightarrow$ reparameterize the α s so that they are *information-orthogonal* to the other parameters in the model (including the β s and ϕ)
- Key idea: Transform the α s so that (for example):

$$E \left(\frac{\partial^2 L_i}{\partial \alpha \partial \beta} \right) = 0$$

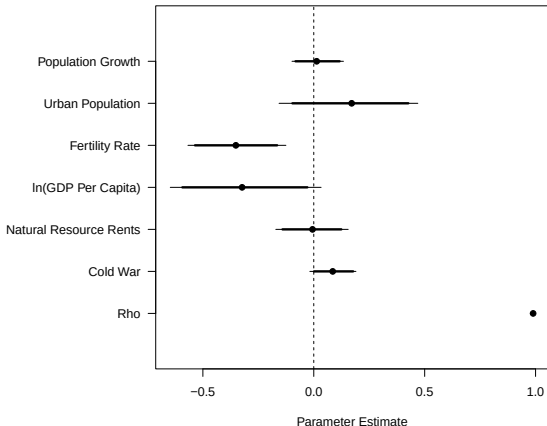
- Can do this via imposition of priors, in a Bayesian framework...
- **In general**, this approach is less assumption-laden and more efficient than the IV/GMM-based approaches discussed above.
- Provides consistent-in- N estimates for T as low as 2...

References:

- Lancaster, T. 2002. "Orthogonal Parameters and Panel Data." *Review of Economic Studies* 69:647-666.
- Pickup et al. (2017) [the "orthogonalized panel model" ("OPM")]

FE + Dynamics Using Orthogonalization

```
> library(OrthoPanels)
> set.seed(7222009)
> OPM.fit <- opm(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+
  lnGDPPerCap+NaturalResourceRents+PostColdWar,data=smol,
  index=c("ISO3","Year"),n.samp=1000)
```



OPM Results: Short- and Long-Run Effects

For $\hat{\phi} \approx 0.98$:

Parameter	Short-Run	Long-Run
Population Growth	0.01	1.07
Urban Population	0.17	13.79
Fertility Rate	-0.35	-30.59
ln(GDP Per Capita)	-0.32	-28.13
Natural Resource Rents	-0.01	-0.45
Post-Cold War	0.09	7.13

What if Y is *trending* over time?

- First Question: Why?
 - Organic growth (e.g., populations)
 - Temporary / short-term factors
 - Covariates...
- Second question: Should we care?
(A: Yes, usually... → “spurious regressions”)
- Third question: What to do?
 - Ignore it...
 - Include a counter / trend term...

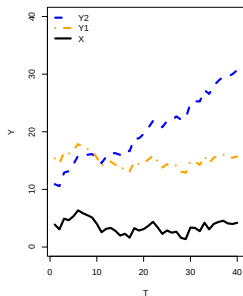
In general, adding a trend term will *decrease* the magnitudes of $\hat{\beta}$...

Trends Matter, Illustrated

Data generating processes:

$$Y_{1t} = 10 + (1 \times X_t) + u_t$$

$$Y_{2t} = 5 + (1 \times X_t) + (0.5 \times T) + u_t$$



	Dependent variable:		
	Y ₁	Y ₂	
X	0.921*** (0.245)	-0.382 (0.786)	0.874*** (0.255)
T			0.482*** (0.026)
Constant	10.320*** (0.917)	20.220*** (2.948)	5.855*** (1.200)
Observations	40	40	40
R ²	0.272	0.006	0.905
Adjusted R ²	0.253	-0.020	0.900
Residual Std. Error	1.802 (df = 38)	5.793 (df = 38)	1.814 (df = 37)

Note:

* p<0.1; ** p<0.05; *** p<0.01

Trends Matter, Part II

Table: FE Models of WBLI

	FE	FE.trend	FE.intx
Population Growth	-0.218 (0.135)	-0.424*** (0.112)	-0.395*** (0.112)
Urban Population	8.047*** (0.437)	1.420*** (0.380)	1.643*** (0.380)
Fertility Rate	-2.920*** (0.292)	3.125*** (0.263)	3.037*** (0.263)
ln(GDP Per Capita)	12.490*** (0.422)	3.247*** (0.383)	2.972*** (0.384)
Natural Resource Rents	1.005*** (0.186)	0.664*** (0.155)	0.656*** (0.154)
Post-Cold War	3.351*** (0.143)	-0.973*** (0.139)	-3.909*** (0.471)
Trend (1950=0)		0.739*** (0.012)	0.674*** (0.016)
Post-Cold War x Trend			0.086*** (0.013)
Observations	8,252	8,252	8,252
R ²	0.520	0.668	0.670
Adjusted R ²	0.508	0.660	0.662

* p<0.1; ** p<0.05; *** p<0.01

“Fixed Effects Individual Slope” models

- Cite: Bruederl, Josef, and Volker Ludwig. 2015. “Fixed-Effects Panel Regression.” In *The Sage Handbook of Regression Analysis and Causal Inference*, Eds. Henning Best and Christof Wolf. Los Angeles: Sage, pp. 327-357.
- FE + unit-level slopes for (some / all) predictor variables
- Equivalent to including $N - 1$ interactions between a predictor $\mathbf{X}$ and each of the α_i s
- Also can test for homogeneity of estimated slopes (Hausman-like test)
- See the `feisr` R package, and its accompanying vignette, or `xtfeis` in Stata

FEIS Example: Post-Cold War

```
> FEIS<-feis(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents | PostColdWar,  
+           data=(smol),id="ID",robust=FALSE)
```

```
> summary(FEIS)
```

Call:

```
feis(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +  
      FertilityRate + lnGDPPerCap + NaturalResourceRents | PostColdWar,  
      data = (smol), id = "ID", robust = FALSE)
```

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-25.6473	-3.3206	-0.0141	3.2288	51.9795

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.431	0.109	-3.96	0.000075 ***
UrbanPopulation	9.088	0.503	18.08	< 2e-16 ***
FertilityRate	-6.697	0.324	-20.67	< 2e-16 ***
lnGDPPerCap	16.651	0.468	35.58	< 2e-16 ***
NaturalResourceRents	0.266	0.162	1.64	0.1

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Normal standard errors

Slope parameters: PostColdWar

Total Sum of Squares: 509000

Residual Sum of Squares: 292000

R-Squared: 0.426

Adj. R-Squared: 0.426

FEIS: Testing

```
> FEIS.test<-feistest(FEIS)
> summary(FEIS.test)
```

```
Call:
feis(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +
      FertilityRate + lnGDPPerCap + NaturalResourceRents | PostColdWar,
      data = (smol), id = "ID", robust = FALSE)
```

Artificial Regression Test

FEIS vs. FE:

H0: FEIS and FE estimates consistent

Alternative H1: FE inconsistent

Model constraints: PopGrowth_hat, UrbanPopulation_hat, FertilityRate_hat, lnGDPPerCap_hat,
NaturalResourceRents_hat = 0

Chi-squared test:

Chisq = 1468.2, df = 5, P(> X2) = 0.0

FE vs. RE:

H0: FE and RE estimates consistent

Alternative H1: RE inconsistent

Model constraints: PopGrowth_mean, UrbanPopulation_mean, FertilityRate_mean, lnGDPPerCap_mean,
NaturalResourceRents_mean, PostColdWar_mean = 0

Chi-squared test:

Chisq = 442.2, df = 6, P(> X2) = 0.0

FEIS vs. RE:

H0: FEIS and RE estimates consistent

Alternative H1: RE inconsistent

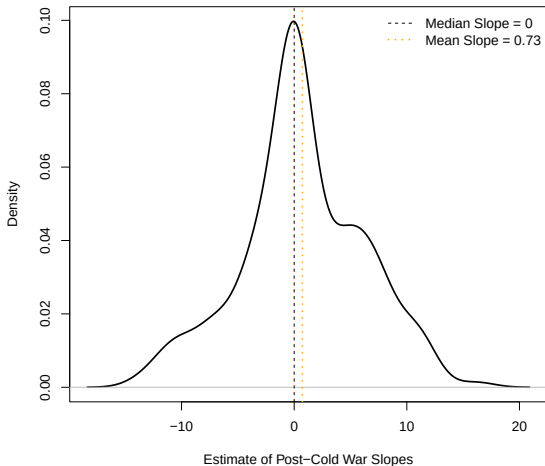
Model constraints: PopGrowth_hat, UrbanPopulation_hat, FertilityRate_hat, lnGDPPerCap_hat,
NaturalResourceRents_hat = 0

Chi-squared test:

Chisq = 1627.1, df = 5, P(> X2) = 0.0

FEIS: Unit-Specific Slopes

Distribution of Unit-Specific Slopes for *Post-Cold War*



FEIS: Unit-Specific Trends

```
> FEIS2<-feis(WomenBusLawIndex~PopGrowth+UrbanPopulation+FertilityRate+lnGDPPerCap+NaturalResourceRents+PostColdWar | Year,  
+ data=(smol),id="ID",robust=FALSE)
```

```
> summary(FEIS2)
```

Call:

```
feis(formula = WomenBusLawIndex ~ PopGrowth + UrbanPopulation +  
      FertilityRate + lnGDPPerCap + NaturalResourceRents + PostColdWar |  
      Year, data = (smol), id = "ID", robust = FALSE)
```

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-18.55664	-2.41355	0.00519	2.37725	45.05731

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
PopGrowth	-0.2904	0.0803	-3.62	0.0003 ***
UrbanPopulation	1.8630	0.6039	3.08	0.0020 **
FertilityRate	0.4718	0.3340	1.41	0.1578
lnGDPPerCap	3.5749	0.5301	6.74	1.7e-11 ***
NaturalResourceRents	-0.0675	0.1201	-0.56	0.5742
PostColdWar	-0.8320	0.1005	-8.28	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Normal standard errors

Slope parameters: Year

Total Sum of Squares: 156000

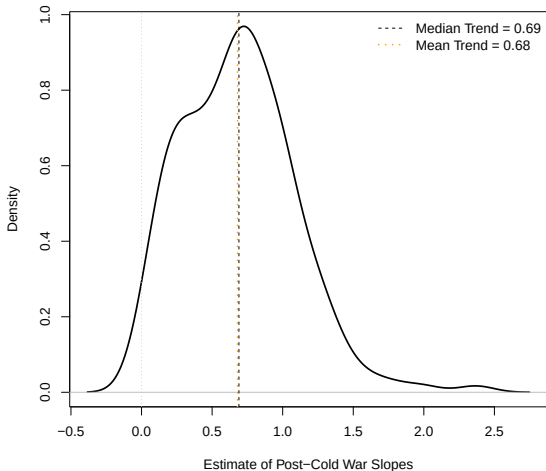
Residual Sum of Squares: 153000

R-Squared: 0.0189

Adj. R-Squared: 0.0182

FEIS: Unit-Specific Trends

Distribution of Unit-Specific Trend Estimates



R :

- the `plm` package (`purtest` for unit roots; `plm` for first-difference models; `pgmm` for Arellano-Bond)
- the `panelvar` package (FD, A&B; Blundell and Bond (FOD))
- the `panelAR` package (GLS-ARMA models)
- the `glS` package (GLS)
- the `pdynmc` package (GMM models via moment conditions)
- the `dynpanel` package (A&H, A&B; minimal...)

Stata :

- `xtgls` (GLS)
- `xtpcse` (PCSEs)
- `xtabond` / `xtabond2` / `xtdpd` / `xtdpdsys` (A&H + A&B + etc. + dynamic models)
- Others...

Final Thoughts: Dynamic Panel Models

Things to consider:

- N vs. T ...
- Are dynamics nuisance or substance?
- What problem(s) do you *really* care about?